

GUEST EDITORIAL

Sustainable industrialisation: the path forward

‘We do not inherit the Earth from our ancestors; We borrow it from our children’. This timeless proverb captures the essence of the environmental challenge confronting the modern era. At a time when economic ambition and ecological fragility coexist in uneasy balance, it is important to decide whether development will proceed in conflict with nature or in harmony with nature.

Economic development and human welfare are inseparable from industrialisation and scientific innovation. Throughout much of the twentieth century, agriculture formed the backbone of India’s economy, contributing the largest share to gross domestic product (GDP) and employing the majority of the population. Today, however, the structure of the economy has transformed. The industrial sector now contributes nearly 30 per cent to the nation’s GDP, while agriculture accounts for approximately 17 per cent. Industry generates large-scale employment, both skilled and unskilled, strengthens infrastructure, boosts trade, stimulates technological progress, and elevates standards of living.

Yet despite its undeniable economic contribution, industry often encounters resistance from communities due to environmental concerns. The problem is not industrialisation itself, but in the way its by-products—sewage, effluents, emissions, and solid waste are managed. India’s enforcement agencies have consistently reinforced the principle that development cannot come at the cost of ecological degradation. They establish a simple but powerful truth: economic growth and environmental protection are not opposing goals; they are complementary responsibilities.

From Linear Industry to Circular Ecology

Nature offers a profound lesson. Every natural process produces both a product and a by-product. In ecosystems, waste from one process becomes input for another, creating seamless cycles: The carbon cycle, nitrogen cycle, and water cycle. Nature operates on circular principles, not linear ones.

Industrial systems, however, historically followed a linear model: extract, produce, consume, discard. During the industrial revolution, the focus was on maximising production and profitability. In the contemporary era, the approach has shifted toward efficiency, reducing water use and energy consumption but the underlying linear structure often remains unchanged. The result has been mounting waste, pollution, and resource depletion.

The urgent need of the hour is to transition to a diversified and circular one, where by-product of one industry becomes the raw material of another. This approach is guided by the 5 R’s: reduce, reuse, recycle, recover, and replenish. In numerous orders of the enforcement agencies concerning river pollution and waste mismanagement, directed states to implement integrated waste management systems and ensure optimal utilisation of sewage treatment capacity. Waste, when intelligently managed, becomes wealth.

Sewage: From Liability to Strategic Resource

Sewage accounts for nearly 70 per cent of water pollution in India. Yet sewage represents not only a liability but also an opportunity. Once consumed, this water becomes sewage. Properly treated, however, wastewater is a valuable resource. Hundreds of MLD of treated water are already being reused to replenish lakes, support industrial processes, irrigate parks, and maintain landscaping around major infrastructure facilities.

Investment in tertiary treatment technologies can produce near-potable water at a fraction of the cost of inter-basin water transfer projects. Treated wastewater acquires tangible market value. In such a scenario, sewage ceases to be an environmental burden and becomes an economic asset. The challenge lies not in technological feasibility, but in scaling up infrastructure, ensuring full utilisation of installed treatment capacity, and building robust sewer networks.

Solid Waste: The Silent Polluter

Urban India generates vast quantities of municipal solid waste every day. While collection efficiency has improved in many cities, scientific processing remains below half of total generation. The unprocessed fraction contributes to methane emissions, groundwater contamination, and river pollution.

The Solid Waste Management Rules, 2016 mandate segregation at source, scientific processing, and the prohibition of unscientific dumping. Technologies such as composting, biomethanation, refuse-derived fuel (RDF) co-processing in cement plants, and waste-to-energy facilities are available and proven. The real impediments are weak enforcement, inadequate monitoring, and insufficient behavioural change at the household level. Cleanliness is not merely a civic virtue; it is a public health imperative.

Polluted River Stretches: A Warning Signal

The Central Pollution Control Board has identified multiple polluted river stretches based on elevated biological oxygen demand (BOD) levels. The principal contributors are untreated sewage and poorly managed municipal waste from urban local bodies. Rivers once revered as sacred lifelines now endure chemical, biological, and physical stress. Elevated BOD, nutrient loading, heavy metals, and faecal contamination threaten aquatic ecosystems and drinking water sources alike. River restoration demands systemic reform in waste management and wastewater treatment.

The Growing Crisis of E-Waste

Electronic waste has emerged as one of the fastest-growing waste streams in the country, driven by rapid technological advancement and rising consumer demand. Discarded computers, mobile phones, appliances, and electronic devices contain valuable materials such as gold, silver, copper, and rare earth elements. Simultaneously, they harbour toxic substances including mercury, cadmium, and lead.

India is now the world's third-largest generator of e-waste. However, only a small fraction is formally collected and recycled. The majority is processed in the informal sector using hazardous methods such as open burning and acid leaching, leading to groundwater contamination, ecological degradation, and loss of recoverable resources. Although E-Waste Management Rules and Extended Producer Responsibility (EPR) frameworks have been introduced, enforcement gaps

and inadequate tracking mechanisms limit their effectiveness. Formalisation of recycling systems and integration of the informal sector are essential to prevent environmental and public health crises.

Redefining Industrial Responsibility

Industries routinely prioritise brand value and customer satisfaction, often recalling defective products at enormous cost. Yet environmental investments are sometimes viewed solely through the lens of short-term economics. Pollution control systems such as ultra-filtration units, reverse osmosis plants, bio-digesters, scrubbers, and electrostatic precipitators are available and technologically proven. The barrier is not innovation, but outlook.

Sustainability is not philanthropy; it is prudent risk management and long-term viability. Communities will embrace industry only when ecological indicators such as air quality, water quality, and soil health are protected or enhanced.

The Way Forward

Expanding sewage treatment plant capacity, ensuring 100 per cent utilisation of installed infrastructure, strengthening sewer networks, enforcing solid waste rules, promoting treated wastewater reuse, encouraging private participation, and adopting circular industrial models are not optional reforms, but they are necessities.

Development detached from ecological responsibility is self-defeating. But industrialisation guided by circular principles, supported by robust regulation, and embraced by responsible citizens can transform sewage into a resource, waste into wealth, and polluted rivers into rejuvenated lifelines.

Industrial growth need not be viewed as a bane. With accountability, innovation, and respect for ecological limits, it can truly become a boon—ensuring that economic progress flows in harmony with environmental sustainability. Only then will development be not merely rapid, but righteous and worthy of the generations from whom we have borrowed this Earth.

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